



Imaging in penetrating neck injuries

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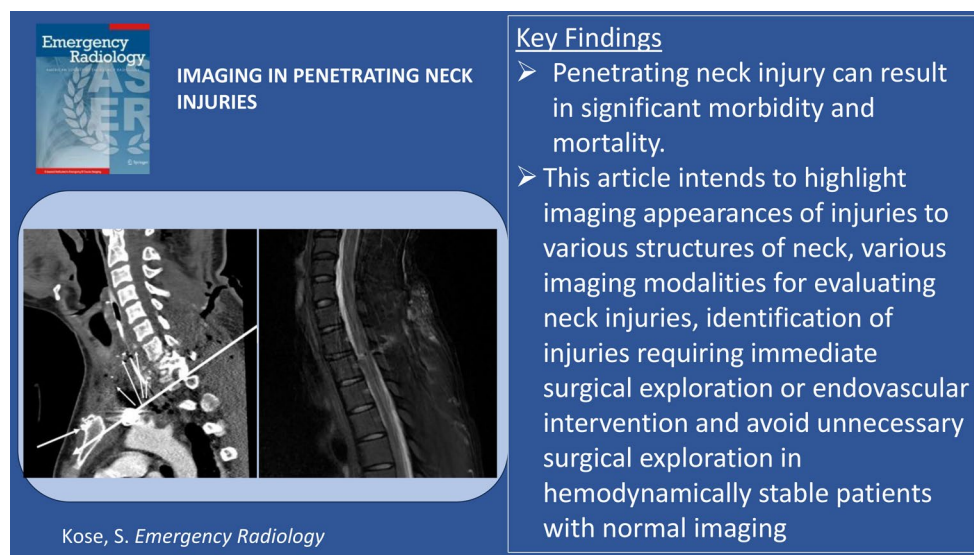
Received: 12 February 2025 / Accepted: 14 April 2025 / Published online: 25 April 2025

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Abstract

The neck is a close compartment wherein many vital structures like carotid and vertebral vessels, larynx, trachea, pharynx, oesophagus and spinal cord are packed closely. As a result penetrating neck injury with either sharp objects like knife or projectile injury can result in significant morbidity and mortality. There have been considerable changes in management strategies of penetrating neck injuries time to time with most of injuries managed conservatively in hemodynamically stable patients. CT angiography plays an important role in detecting injuries to structure which require immediate surgery/endovascular intervention and helps to avoid unnecessary surgical exploration in stable patients. This article intends to familiarize radiologists with imaging features of injuries to various structures of neck which shouldn't be missed and highlight stepwise approach for evaluation of neck structures in penetrating injury.

Graphical Abstract



Keywords Penetrating neck injuries · Trajectory · Bullet · Arterial injuries · Laryngeal injuries · Spinal cord injuries

Introduction

Penetrating neck injuries occur in ~1–10% individuals presenting in trauma settings [1, 2] with high rate of mortality of ~2–10% owing to tightly packed vital structures like carotid, vertebral arteries, aerodigestive tract and spinal cord within small anatomic compartment [3–6]. There has been considerable shift in the management strategy of

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patient with penetrating neck injuries from mandatory surgical exploration of all penetrating injuries which was practiced earlier to conservative management in most of patients based on clinical and imaging findings. This article intends to highlight imaging appearances of injuries to various structures of neck, various imaging modalities for evaluating neck injuries, identification of injuries requiring immediate surgical exploration or endovascular intervention and avoid unnecessary surgical exploration in hemodynamically stable patients with normal imaging.

Evolution of management of penetrating neck injuries

Earlier, from the era of civilian war to World War II, most of the penetrating neck injuries were managed conservatively [7–10]. However, by the end of World War II, it was realised that penetrating neck injuries resulted in significant morbidity and mortality mainly due to stroke and neuromuscular compromise which are missed on physical examination [6, 11–17]. After World War II to last few decades, almost all penetrating neck injuries with breach to platysma were managed with surgical exploration [6, 15, 18–22]. Most of the explorations did not yield positive findings and it was realised most patients can be managed conservatively [3–5, 20].

Later, zone based approach for management was developed based on which injuries were managed based on zone of entry wound and depth of penetration. Neck was divided into 3 zones. The boundaries and important contents of various zones are described in Table 1.

According to this approach, all injuries in zone II underwent surgical exploration. Zone I and zone III were difficult to access by surgery and surgery was performed in only selected cases and injuries in these zones were evaluated by different modalities like conventional angiography, esophagography, endoscopy, ultrasound and other modalities [18–22]. MDCT angiography is a recently developed investigation for evaluation of PNI and is fast, accurate and reliable technique in excluding injuries requiring prompt surgical exploration or endovascular intervention [6, 10, 13, 15, 16, 23, 24].

Recently, penetrating neck injuries are managed based on clinical findings. Hard and soft signs predicting injuries to various vital structures are shown in Table 2. According to this approach, patients with hard signs on physical examination should undergo immediate surgery without need for imaging [2, 25–27]. While patients with soft signs should undergo MDCT angiography, further management of these patients depends on MDCT angiography findings.

Table 1 Zones of neck and their contents

Zones of neck	Lower Border	Upper border	Vital structures
Zone I	Upper border of both clavicles and sternal notch	Cricoid cartilage	-Brachiocephalic veins -Innominate artery -Common carotid, proximal subclavian and vertebral arteries -Internal jugular veins -Trachea -Esophagus -Thyroid -Cervical spine and spinal cord -Sometimes, lung apices and aortic arch
Zone II	Cricoid cartilage	Angle of mandible	-Common carotid, proximal internal and external carotid arteries -Vertebral arteries -Internal jugular veins -Larynx -Pharynx -Cervical spine and spinal cord
Zone III	Angle of mandible	Skull base	-Internal and external carotid arteries -Vertebral arteries -Internal jugular veins -Cervical spine and spinal cord

Table 2 Hard and soft signs of injuries to vital structures of neck

Structure	Hard signs	Soft signs
Arteries	Pulsatile active haemorrhage Pulsatile/ expanding hematoma Palpable thrill/ audible bruit Pallor Pulselessness Parasthesias Shock	History of pulsatile bleed at the site on injury Non-expanding hematoma Hypotension
Aerodigestive tract	Airway compromise Subcutaneous emphysema Hemoptysis Hematemesis Stridor Hoarseness Air bubbling through the wound	Dysphonia Dysphagia
Nervous system	Stroke Neurological deficit	Neurological deficit

Multiple studies have shown CTA to have high sensitivity, in the range of 90–100%; specificity ranging from 98.6 to 100%; a positive predictive value of 92.8–100%; and a negative predictive value of 98–100% [28–30] for evaluating vascular injury. In addition to identifying vascular injury, CTA simultaneously identifies extravascular soft-tissue and aerodigestive injuries with a sensitivity of 100% and a specificity ranging from 93.5 to 97.5% [2, 25, 26, 28–32].

Mechanism of injury

Penetrating injuries are classified as either low energy or high energy injuries. Low energy injuries include low velocity projectile injuries like handgun ammunition and sharp objects like knives, nail, etc. High energy injuries include high velocity bullet injury from military rifles.

Injuries due to projectile are most commonly due to stretching and crush injury [33–35]. Harvey and Korr [35] described two types of tissue damaging pressure waves produced by projectiles such as bullets: First, sonic shock wave is produced before the projectile hits the target. This sonic shock wave travels with speed of sound and produces no cavitation. Secondary pressure wave is produced when projectile hits the target, deforms tissue and radiates laterally leading to transfer of kinetic energy to surrounding tissues and cavitation.

The bullet may fragment into small pieces or cause bone fragmentation. These bone and bullet fragments can serve as secondary projectiles and produce further damage [36].

The severity of injuries caused by high velocity bullets depend on degree of deformation of target tissues, fragmentation of bullet, velocity and size of of projectile [33]. According to Hague convention of 1899, most military rifle bullets are fully encased in metal jacket which decreases its deformation and fragmentation capacity, such weapons are designed to wound the target and not kill them. Most of ammunition used in homicide do not obey Hague rules and produce greater injury [33, 35].

Pellets from shotguns are usually smaller in size and have low velocity and hence when shot transfer almost entire kinetic injury to surrounding tissues and may not exit, leading to higher tissue damage. Severity of injury depends on distance from the target, number of pellets that hit the target, size and pattern of the pellets that strike the target.

Evaluation of wound trajectory

The zone of injury depends on site of entry wound [7, 8, 37]. However, track of wound may traverse from one zone to other and determining trajectory of penetrating injuries is important to detect potential structures likely injured, so that appropriate further investigations and management can be done [13, 36, 38–40].

In high velocity projectile injuries, the track can be traced by bullet or beam hardening artefacts that occur in vicinity of bullet in the absence of deviation of bullet path.

On striking bone, the bullet may fragment into multiple pieces and also fracture bones resulting in multiple bullet/bone fragments which can serve as secondary projectiles. Usually, these traverse in the same direction as bullet, however sometimes they may deviate from original path into various directions causing injury to surrounding structures (36). In such cases, trajectory of individual fragment needs to be traced (Fig. 1).

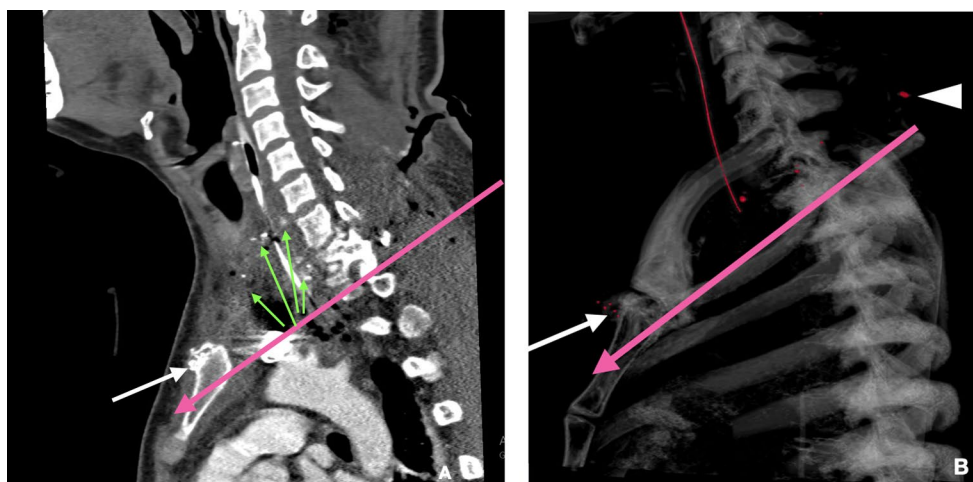
In low velocity injuries by handgun, tracing track is difficult [41] as entry wound may not have exit wound and smaller diameter of track.

In case of neck injury by sharp objects like knives, trajectory can be known by direct evidence of weapon left in situ which is not encountered in most circumstances. In such cases, trajectory can be estimated by the presence of air foci and areas of haemorrhage within the tract. Alternatively, a straight line can be drawn connecting the two structures showing direct evidence of injury [38].

Role of imaging in penetrating neck injuries

CT angiography is the current workhorse for deciding management in patients with penetrating neck injury who are hemodynamically stable or have soft signs of injury to neck structures. The various modalities used in evaluation of

Fig. 1 Trajectory of gun shot injury CT angiography image of neck (A) reveals entry wound on back of neck (arrowhead) with trajectory indicated by linear air foci, traversing anteroinferiorly (pink arrow) and exit wound at anterior border of manubrium sterni (white arrow). Note bone and bullet fragments have different trajectories serving as secondary projectiles (green arrows) VRT image (B) better depicts track of primary bullet with entry wound on back of neck and exit wound at manubrium sterni (white arrow)



neck injuries, their indications, advantages and limitations are summarised in Table 3.

Injuries to vital structures of neck

Cervical vascular injuries

Arterial injuries

Cervical arterial injuries occur in ~15–25% of patients [16, 19, 23]. Injuries to carotid arteries are twice as common as vertebral arteries with carotid vessels involved in 80% of cases and vertebral arteries in 43% of cases [38]. Carotid injuries are devastating because of increased incidence of stroke in 15% cases and death in 22% cases [16, 17, 42].

Direct signs of vascular injuries include partial or complete thrombosis (Fig. 2), intimal injury, dissection, pseudoaneurysm (Fig. 3), arteriovenous fistula, extravasation of contrast (Figs. 4 and 5) and transection [38]. Although direct injuries may not be visible on CT angiography, arterial injuries should be suspected in presence of periarterial hematoma/ fat stranding and carotid arteries lying along course of wound track or bone/ missile fragments/ air foci within 5 mm of arteries [38].

In patients with normal CT findings and high clinical suspicion of vascular injury or equivocal CT findings, conventional angiography may be indicated to exclude vascular injuries [38]. Management of patients with vascular thrombosis depends on neurological status and imaging findings of brain. For example, in patients diagnosed with a carotid artery occlusion, a significant neurological deficit, and a large cerebral infarct on the ipsilateral side, observation is the preferred treatment of choice due to the poor prognosis in this subgroup of patients. Similarly, in patients with carotid artery occlusion and a normal neurological examination, observation and anticoagulation therapy for a period of at least 3 months is recommended to avoid any propagation of existent thrombus.

Minimal vascular injuries do not require as aggressive surgical approach. Minimal injuries can be described as nonobstructive or adherent intimal flaps and pseudoaneurysms less than 5 mm in size. Arterial repair is always preferred over ligation of carotid arteries [43, 44]. Repair can be performed by lateral suture patch angioplasty, end to end anastomosis, interposition or bypass grafting. Endovascular approach with covered stents can be used for treatment of small arteriovenous fistulas, active contrast extravasation and pseudo aneurysms.

Single vertebral artery injuries rarely lead to neurological deficits and can be managed conservatively or by endovascular approach. Surgery is performed only in cases where

active bleed from vertebral arteries is present or angiographic methods of treatment have failed.

Venous injuries

Injuries to internal and external jugular veins are usually missed on physical examination and are clinically insignificant. Venous injuries occur in ~16–18% cases with penetrating neck trauma [18, 28, 45] and when present should be mentioned in report. Direct signs of injury include thrombosis, intimal injury and focal defect with active contrast extravasation. The low-pressure venous system usually tamponades or occludes without a major hemorrhage or hematoma [46]. Venous injuries are usually detected when surgical exploration is performed for repairing other vital structures. An injury to the internal jugular vein should be repaired by a lateral venorrhaphy [47]. However, when the patient is in shock or repair is difficult, any venous injury should be managed by ligation. The external jugular vein can be ligated without adverse sequelae.

Aerodigestive tract

Larynx

Injuries to larynx should be reported in terms of location (supraglottic, glottic or subglottic), structures injured (cartilage, mucosa, ligaments, joints, nerves) and severity of injury. Whenever defect in laryngeal cartilage is detected (Figs. 6 and 7), displaced of cartilages and airway compromise needs to be reported [7, 31, 48–50]. Undisplaced stable fracture of cartilage is usually managed conservatively while displaced cartilage fracture requires surgical management [51]. Status of vocal cords must be reported to look for signs of avulsion or recurrent laryngeal nerve injury. Many times, defect in walls of larynx obscured by surrounding hematoma. In such cases, laryngoscopy needs to be performed to exclude injury when larynx lies along course of wound track.

Trachea

Tracheolaryngeal injuries occur in 1–7% of patients with penetrating neck trauma [40, 48, 52]. Direct signs of defect in tracheal wall are not often detected on imaging. Tracheal injury should be suspected in cases with extensive subcutaneous emphysema. When trachea lies along course of wound track, endoscopy should be done to exclude tracheobronchial injury [3, 10, 12, 40]. Complete or incomplete laceration of tracheal wall needs to be reported as incomplete laceration of trachea with opposed ends are usually

Table 3 Imaging modalities in evaluation of penetrating neck injuries

Modality	Uses	Advantages	Disadvantages
Radiography	-For detecting Vertebral fractures -detecting presence of metallic foreign bodies -Evaluation of airway patency and subcutaneous emphysema	-Cheap -Easily available -Portable radiographs available	-Soft tissue, vascular injuries cannot be evaluated
Ultrasound	-Detecting of patency of cervical arteries and veins -Evaluation of flow characteristics and velocity using doppler -Associated injuries to muscles and soft tissues	-Easily available -Cost-effective	- Operator dependent - Laryngotracheal and pharyngoesophageal injuries cannot be assessed - Cervical spine and spinal cord injuries cannot be assessed - Inadequate visualisation in areas of bandages, subcutaneous emphysema and open wounds - Inadequate visualisation of vertebral arteries, zone I and III injuries
MDCT angiography	-Currently the primary workhorse technique in penetrating neck injuries with soft signs -Evaluation of cervical vascular as well as non vascular structures like aerodigestive tract, cervical spine and soft tissues	-Fast, accurate and reliable -Vascular and non-vascular injuries can be evaluated simultaneously -Avoids unnecessary therapeutic surgical exploration	-Streak artefacts from metallic foreign bodies may obscure or simulate vascular injury -Increased contrast dose to the patient if vascular injury warrants endovascular intervention for which catheter angiography needs to be performed -Technical errors like late commencement of arterial phase may lead to venous contamination, as a result arteriovenous fistulas may be missed -Body habitus reduces image quality especially in lower neck and upper thorax -Motion artifacts may obscure/ simulate vascular injury
MRI	-Evaluation of spinal cord, traumatic disk injury, ligamentous injury and laryngeal cartilaginous injuries	-Injuries to soft tissues can be better evaluated due to high soft tissue contrast resolution	-Presence of metallic foreign bodies precludes use of MRI -Osseous injury not well evaluated -Not easily available -Costly
Conventional angiography	-Zone I and zone III injuries with high grade of clinical suspicion for vascular injuries -Inadequate assessment of vascular structures due to streak artefacts from metallic fragments lying close to vessels -CT angiography findings are equivocal	-Detection of vascular injuries missed on CT angiography -Endovascular intervention can be performed in the same setting	-Non-vascular injuries cannot be evaluated -Complications like puncture site hematoma, access vessel thrombosis, thromboembolization distal to the access point, arteriospasm, ischemia, and arterial dissection
Endoscopy	-Evaluation of suspected laryngeal, tracheal and esophageal injuries	- Confirmation of findings on CT angiography - Detection of injuries not detected on CT angiography	- Injuries not involving mucosa can be missed - Vascular other soft tissue injuries cannot be evaluated
Esophagography	-Esophageal / pharyngeal injuries suspected on CT angiography	- Leaks from pharynx and oesophagus better seen	- Pharyngeal injuries can be missed - Adjacent hematoma can mask the defect

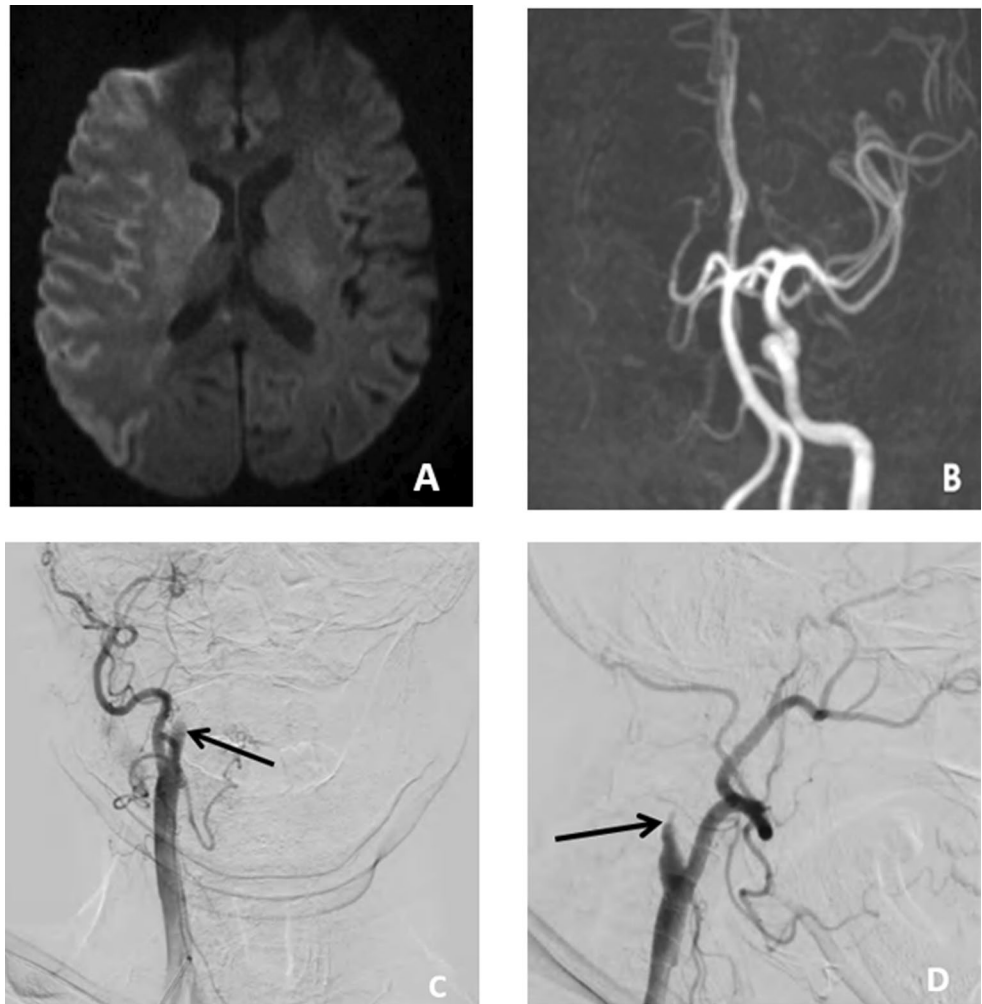
managed non-operatively with endotracheal intubation while complete laceration of tracheal wall needs surgical management [51].

Pharynx and oesophagus

Esophageal injuries are very rare and occur in ~0.9–6.6% cases of penetrating neck trauma [7, 40, 53–55]. Most common cause of death in patients with esophageal injuries include delayed detecting leading to mediastinitis and sepsis [38]. Hence, injuries involving pharynx and oesophagus

need to diagnosed and treated immediately. Injuries to pharynx and oesophagus should be suspected when wound track appears to traverse along course of oesophagus (Fig. 8), bone fragments/metal fragments lie close to these structures, surrounding hematoma/ air foci in vicinity of oesophagus are present. When these signs are present on CT angiography, conventional/ CT esophagography should be done to rule out esophageal injuries. In patients with equivocal findings and suspected pharyngeal injuries, endoscopy must be performed as pharyngeal injuries are usually missed on esophagography. Combined sensitivity of esophagography

Fig. 2 Internal carotid artery thrombosis - MRI diffusion weighted image (A) in a case of stab injury to right side of neck with left sided weakness reveals acute infarct in right middle cerebral artery territory. MRI time of flight angiography image (B) shows occlusion of right internal cerebral artery (ICA) and middle cerebral artery. Digital subtraction angiography images (C and D) showed occlusion of right ICA from origin (black arrow)



and endoscopy in detecting pharyngo-esophageal injuries is ~90–100%.

Esophagography is usually a biphasic procedure with the initial study performed using oral water soluble contrast medium and second step includes administration of barium if no leak is observed on administration of water soluble oral contrast. However, monophasic study with water soluble contrast has proved to be equally effective [53].

Whenever oesophageal perforation is seen on esophagogram, it should be mentioned if it is contained/ uncontained leak as contained leaks are usually managed conservatively and may need follow up imaging after few weeks while uncontained perforation needs repair. A contained leak is best described as extravasation of contrast on esophagogram, but the contrast “loops back in” after initially extravasating from the esophagus. Degree of inflammation or soiling

of surrounding soft tissue also needs to be mentioned as patients with greater degree of soiling need insertion of drainage tube and delayed repair for inflammation to subside which may interfere with healing of repaired segment of oesophagus.

Cervical spine and spinal cord

Cervical spine injuries occur in 11–14% cases of penetrating neck trauma [56–59]. High velocity projectiles like bullets from military rifles cause fracture of vertebrae with fragmentation of bones. These bone and bullet fragments can travel along the same direction as bullet or deviate in different directions. Low velocity injuries from sharp objects usually result in fracture of vertebrae without bone fragmentation.

Fig. 3 Common carotid artery pseudoaneurysm - Anteroposterior (A) and lateral (B) Neck Xray of patient with accidental gunshot injury to neck shows gun pellet in soft tissue in left side of neck (white arrow). CT angiography neck (C) shows pseudoaneurysm (black arrow) arising from posterior wall of common carotid artery anterior to pellet (white arrow). Digital subtraction angiography images (D and E) show pseudoaneurysm (black arrow) arising from posterior wall of common carotid artery anterior to pellet (white arrow)

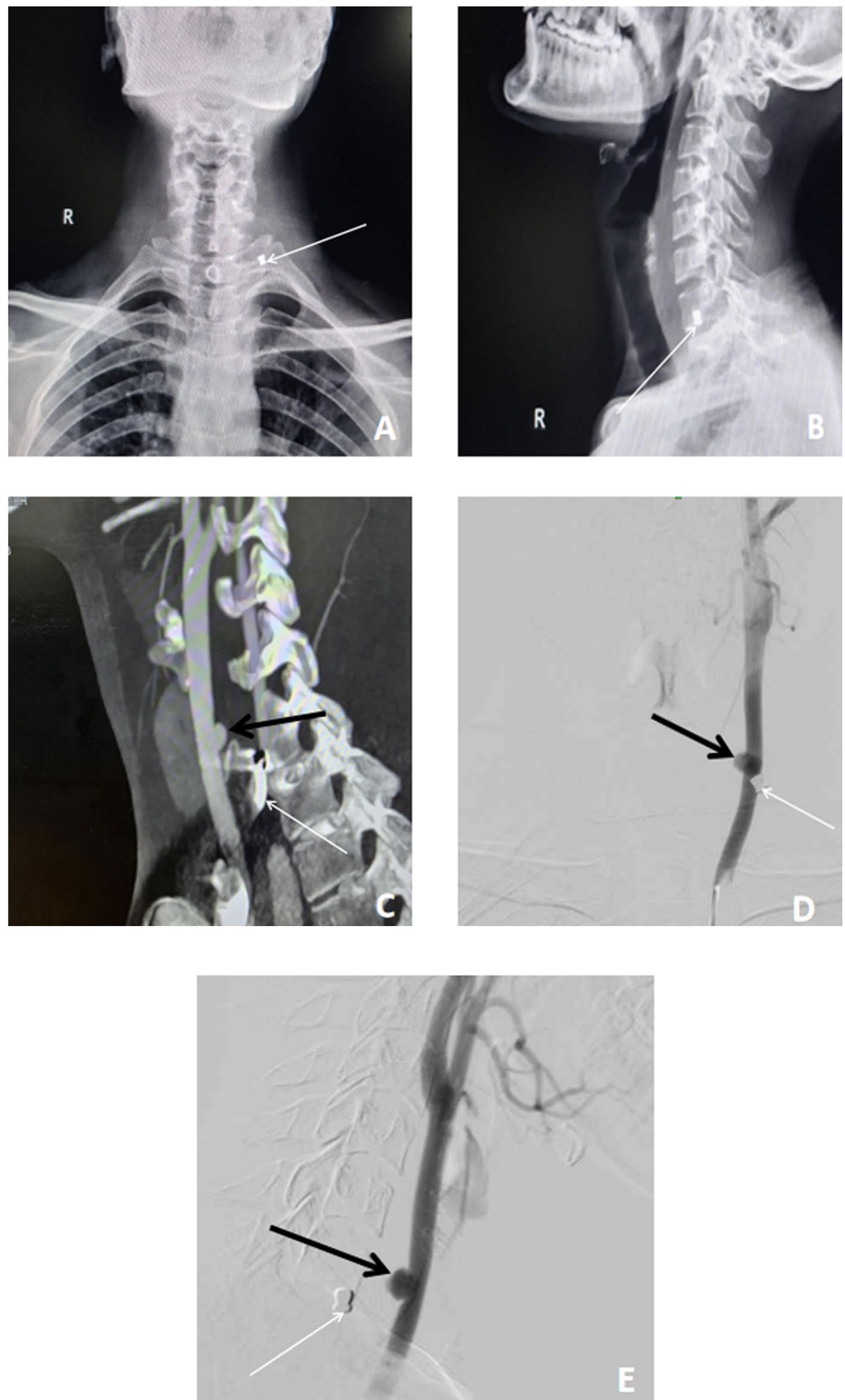


Fig. 4 Vascular injury - CT angiography of neck with stab injury to left side of neck shows hematoma in left sternocleidomastoid muscle with focal area of high attenuation within left sternocleidomastoid muscle (white arrow). On venous phase, there is increase in area of hyperattenuation (white arrow) s/o active contrast extravasation likely due to injury to small muscular branches

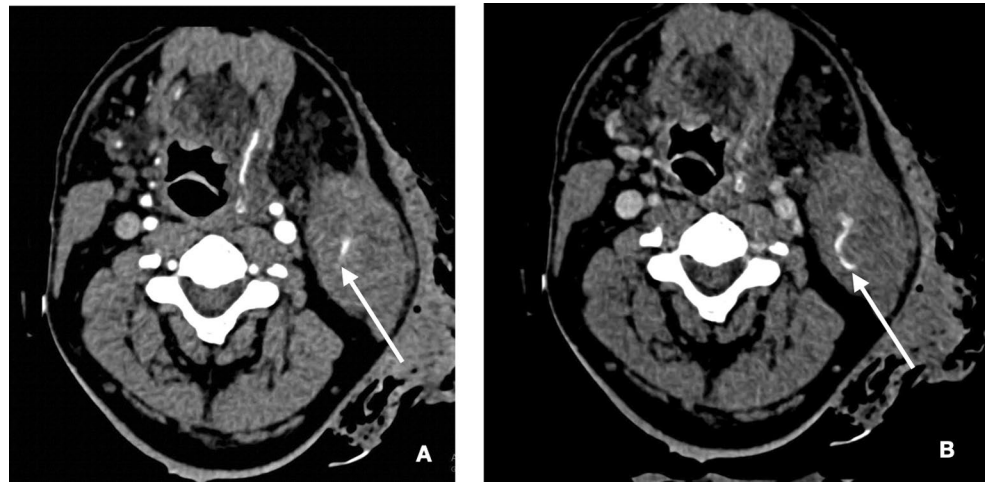
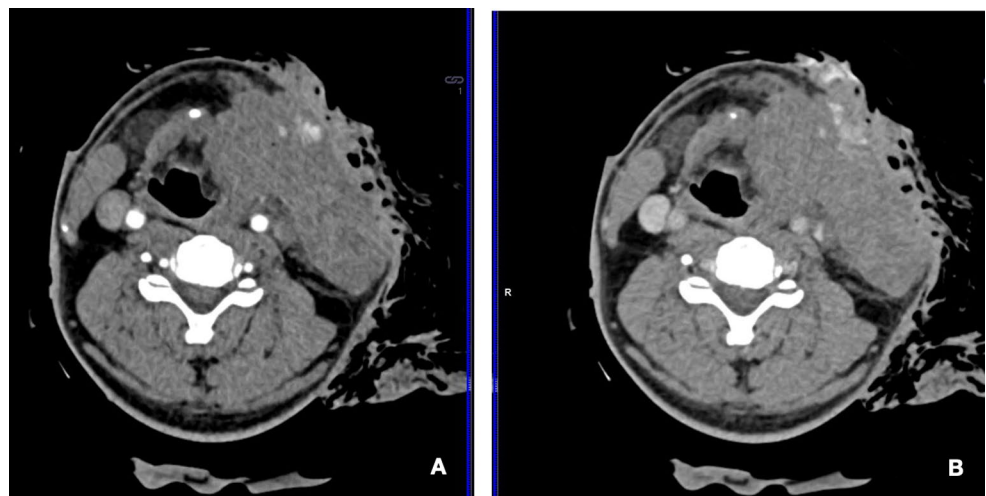


Fig. 5 Vascular injury - CT angiography of neck in arterial (A) and venous phase (B) in patient with stab injury to left side of neck reveals laceration of skin, subcutaneous tissue and left sternocleidomastoid muscle with associated hematoma and active contrast extravasation likely due to injury to muscular branches



Cervical spinal cord injuries are not directly detected on CT. They should be suspected when wound tract extends across the spinal cord or bone/missile fragments are seen within spinal canal (Fig. 9). Cord injuries can also occur due to secondary cavitation of cord from bullet injury [60].

Location and distribution of bone and missile fragments, any signs of dural violation like air, missile/ bone fragments within spinal canal should be noted.

MRI can be performed for confirmation and characterisation of cord injuries (Fig. 10) provided no contraindications for entering MR suite are present (i.e. no metallic foreign bodies).

Stepwise approach for diagnosis and management of patients with penetrating neck injury is summarised in Fig. 11.

Fig. 6 Cricoid cartilage injury - Axial contrast enhanced CT scan of neck in a patient with stab injury to left side of neck (zone I/ II) reveals laceration of skin, subcutaneous tissue and left sternocleidomastoid muscles with associated hematoma in left sternocleidomastoid muscle (arrowhead). Defect in cricoid cartilage seen on left side (arrow) with air foci adjacent to the defect. Note normal cricoid cartilage on right side

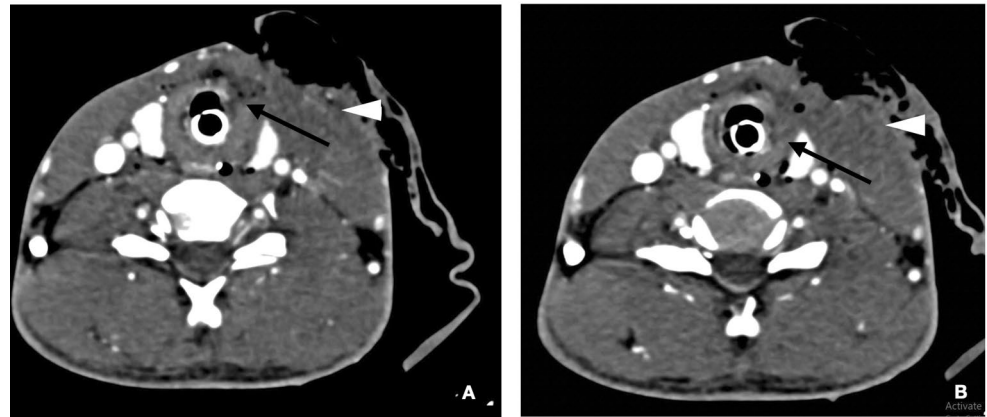


Fig. 7 Thyroid cartilage injury - Axial CT angiography of neck (A and B) in a patient with stab injury to left side of neck (zone II) reveals laceration of skin, subcutaneous tissue and left sternocleidomastoid muscles with associated hematoma in left sternocleidomastoid muscle (arrowhead). Defect in left lamina of thyroid cartilage seen (arrow) with air foci adjacent to the defect. CT angiography in another patient with stab injury to left side of neck reveals defect in left lamina of thyroid cartilage (broken arrow) with associated hematoma in adjacent left sternocleidomastoid muscle and active contrast extravasation (pink arrow). No displacement of fractured ends seen

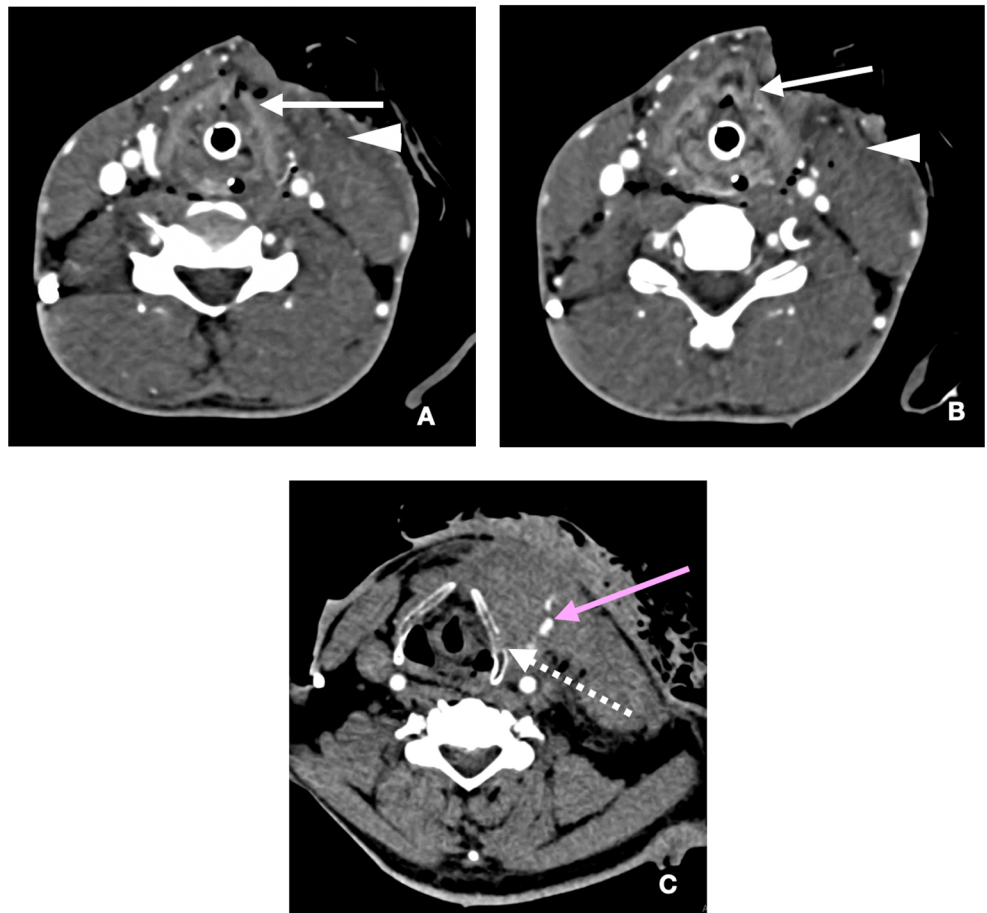


Fig. 8 Cervical esophagus injury - CT angiography of neck in a patient with gun shot injury to back of neck shows trajectory of bullet as linear arrangement of air foci, fracture of spinous process, inferior endplate of C7 and superior endplate of T1 vertebra with bone fragments at posterior aspect of oesophagus (arrow) and bullet fragment at anterior aspect (broken arrow) suggestive of cervical oesophagus along track of bullet fragments. CT esophagogram however, did not show any leak in this patient

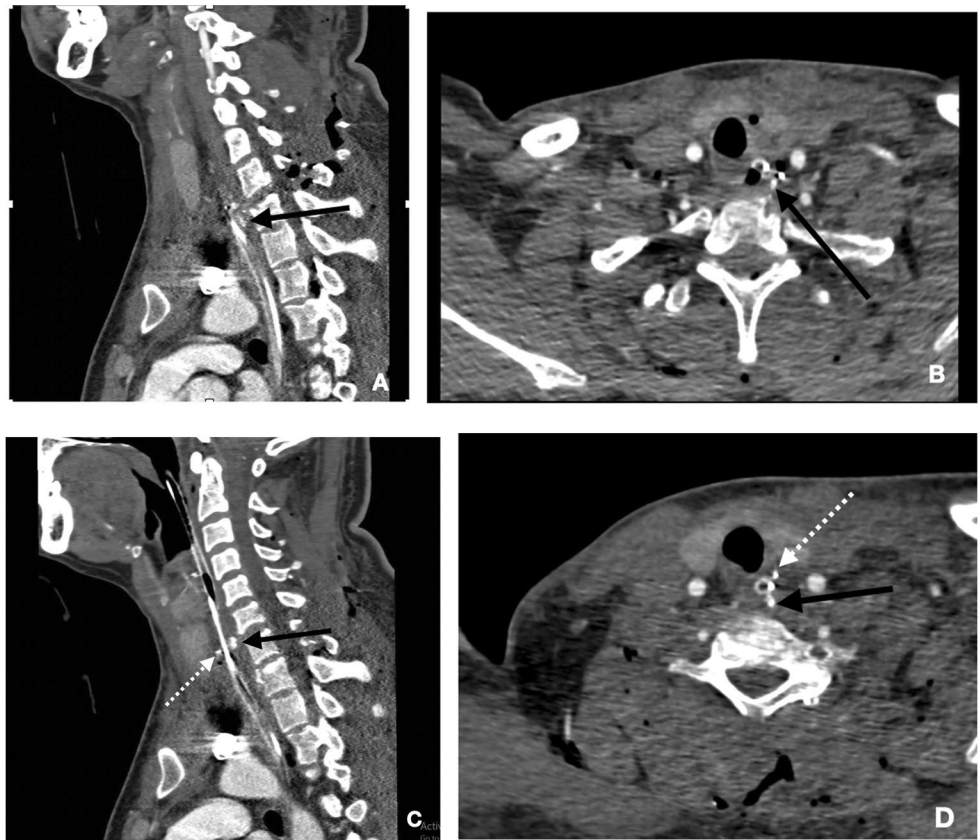


Fig. 9 Cervical spine injury - CT of neck in patient with history of gun shot wound to back side of neck (zone II) reveals trajectory of bullet as indicated by linear arrangement of air foci, fracture of spinous process (white arrow), inferior end plate of C7 vertebra and superior end plate of T1 vertebra (arrowhead) along track of bullet. Few bullet fragments are impacted within inferior end plate of C7 vertebra (broken arrow). Bone fragments are seen within left C7-T1 neural foramen (thin arrow)

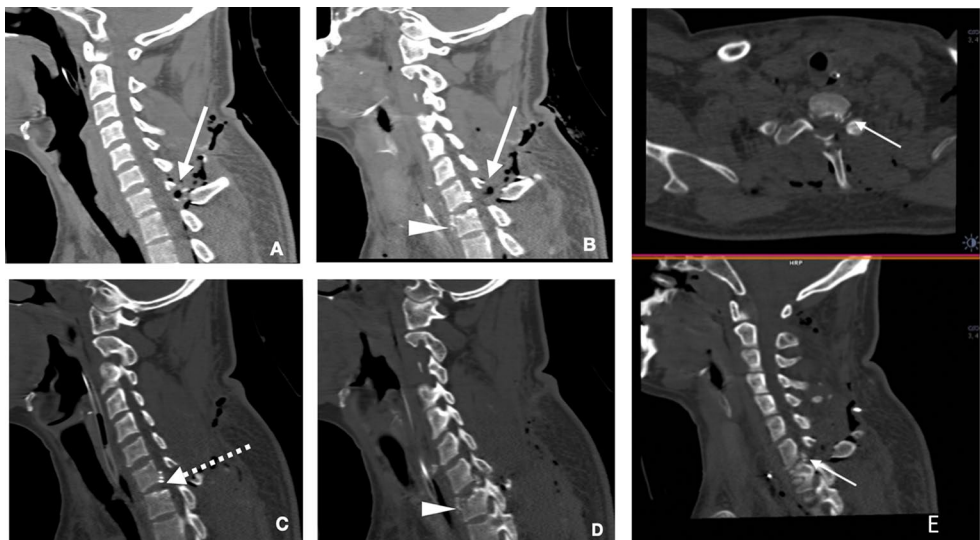


Fig. 10 Cervical spine cord injury - NCCT neck image (A) of a patient with stab injury to neck reveals fracture of left lamina of D2 vertebra (thin arrow). Sagittal T2 fat saturation image (B) reveals T2 hyperintense tract traversing from skin of nape of neck to spinal cord causing spinal cord laceration (thick arrow). Axial T2WI and MERGE sequences (C) show hypointensity within spinal cord s/o haemorrhage (arrowhead)

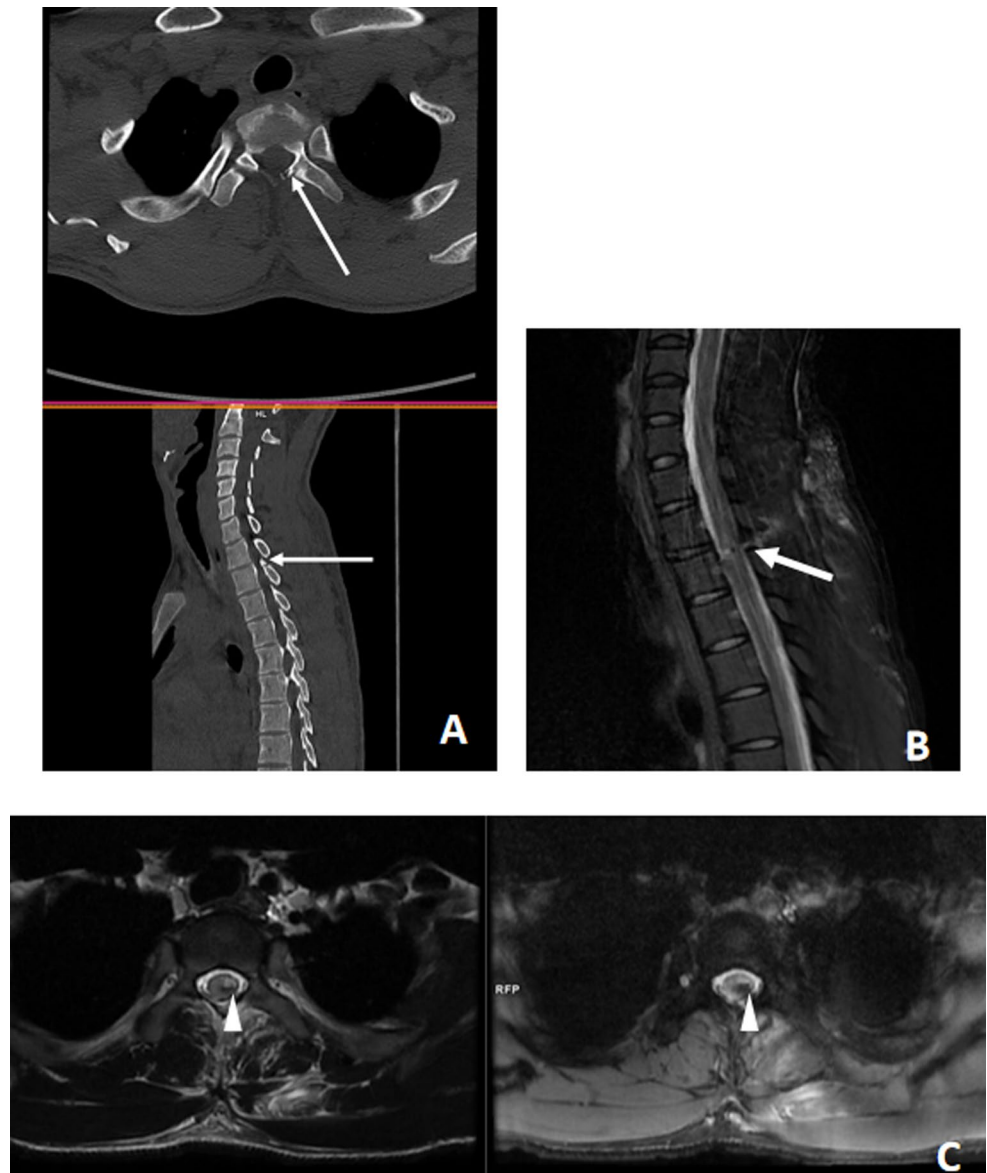
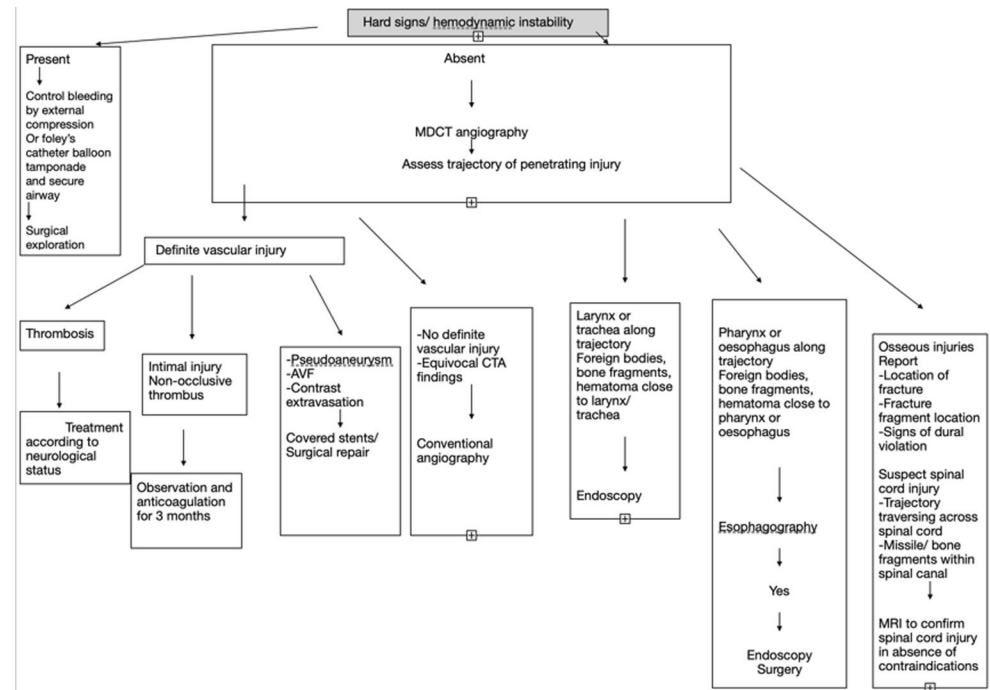


Fig. 11 Stepwise approach for diagnosis and management of patients with penetrating neck injury



Conclusion

Penetrating neck injuries are major cause of morbidity and mortality in trauma settings as many vital structures are packed closely in small compartment. CT angiography plays a vital role in detecting injuries requiring immediate treatment and avoids unnecessary surgical exploration in stable patients. Timely detection of injuries helps in early treatment and reduction of both morbidity and mortality associated with penetrating neck trauma.

Data availability All figures, tables and data for preparing manuscript are available in main paper.

Declarations

Conflict of interest The authors declare no conflict of interest.

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